Multi-Core Sigmoid (MCS)

A sigmoid implied-variance base with zero-wing hats for WW and event smiles

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Abstract

This note documents the Vol-Fitter's sigmoid family, the *Multi-Core Sigmoid* (MCS) model. The base is a six-parameter sigmoid implied-variance slice: a log-cosh convexity primitive giving a variance level and slope *at its fitted centre*, a convexity amplitude, and *asymmetric* put/call wing steepnesses, C^2 across the centre. On top of the base the model adds R signed *zero-wing hat* kernels — normalized centred second differences of the log-cosh primitive — each of which inserts a local variance hump ($\alpha > 0$) or notch ($\alpha < 0$) *without moving the base wing slopes*: the hats are asymptotic-wing-neutral, which is weaker than Lee-compatible — admissibility of the preserved base wings is a separate, diagnosed condition. This local-correction structure is exactly what a single hyperbola (SVI, Note 02) cannot provide: we prove that a globally convex slice cannot reproduce a WW-shaped smile (central trough with two shoulders), and show that two hats reduce the fit error on such a smile from 153.05 to 2.55 volatility basis points. The model carries $6 + 4R$ parameters, has a closed-form Durrleman butterfly diagnostic, and is calibrated in three stages (base $\rightarrow$ greedy hat seeding $\rightarrow$ joint bounded refine) with an analytic Jacobian measured 3.38$\times$ faster than finite differences on the final-refine microbenchmark. The cores count R is the user's "cores" slider — the direct analogue of the LQD Legendre order — capped at two, because the offline backtest shows a third core buys almost nothing out-of-sample at several times the cost while worsening wing arbitrage; a default-on put-wing Durrleman regularizer that vanishes on any clean slice, and a note on that over-fit tendency, close the discussion.

Contents

Conventions, and what you control

Conventions. $k = \log(K/F)$ is log-moneyness; $v = \sigma_{\text{imp}}^2$ is *annualized implied variance* (the quantity MCS models); $w = \tau v$ is total variance, where τ is the production *variance clock* — calendar time dilated by scheduled events (Notes 00, 11), equal to the calendar year fraction t only when the event clock is off. Production passes τ (`prepared.tau`) to the calibrator; the formulas below write a bare t, and wherever it plays a time role, read τ. The z-scale reference σ_{ref} is the *quoted* volatility nearest the money, with a median-of-quotes fallback when that quote is degenerate ($\leq 0.1\%$); it is fixed before the fit, never optimized. A *vol bp* is 10^{-4} in volatility.

What you control, and what you observe. For a desk user: MCS is the `sigmoid` family in the model selector; LQD remains the routine default slice model, and MCS is the event/WW specialist. If MCS is selected the product runs $R = 2$ cores with the put-wing penalty at 100%; the *library* defaults are $R = 0$ and penalty off (research code keeps the full family). The user controls three things: the cores slider $R \in \{0, 1, 2\}$, the amplitude ridge, and the wing-penalty strength. Individual hat parameters $(\alpha_r, c_r, h_r, \kappa_r)$ are *not* trader handles — they are non-unique by design and nothing downstream consumes them. What you observe is the same as for every model: the fitted curve, the full-curve ATM level/skew/curvature handles (computed from the displayed curve — note the base parameters V_0, S_0 are *not* these handles, and the hats can move them), and the per-fit quality diagnostics including g.

1 Motivation: what a hyperbola cannot do

SVI's strength — one convex hyperbola — is also its ceiling. Some smiles are not convex. Ahead of a binary event, or in names with bimodal positioning, the implied-volatility curve develops a *WW* shape: a trough near the money, a *shoulder* (local maximum) on each side, and then wings that turn up again toward the Lee asymptotes. Figure 1 shows such a target; the base sigmoid alone (the dotted curve) misses the shoulders by 153.05 vol bps.

Proposition 1 (A convex slice cannot make a WW)**.** *If total variance $w(k)$ is convex on an interval, then on that interval w has no interior strict local maximum, hence no shoulder flanked by two troughs. A WW smile, which has interior local maxima, therefore cannot be represented by any globally convex slice (in particular by raw SVI, whose $w'' > 0$ everywhere).*

Proof. A convex function on an interval lies below the chord between any two points and its one-sided derivatives are non-decreasing; an interior strict local maximum would force the derivative to change from positive to negative, contradicting monotone non-decreasing slope. SVI has $w''(k) = b\sigma^2/r^3 > 0$ (Note 02), so it is strictly convex. □

The MCS answer is to keep a convex *base* that owns the wings and the overall skew, and to add *local, wing-neutral* corrections that can be positive or negative.

> **Invariant**
>
> 1. Adding cores never moves the asymptotic wings: every hat and its first two derivatives vanish in both tails (lemma 1).
>
> 2. Expressiveness is budgeted: cores are capped by the quote count (10) and the surfaced slider

is clamped to two.

3. The model reports its own butterfly diagnostic g per fit, and the put-wing hinge rows are exactly zero on any clean slice — penalty *off* is locked byte-identical; penalty *on* over a clean slice is locked unchanged to solver tolerance (10^{-9} relative), test-locked either way.

4. The parameters are a means, not the object: hats are non-unique by design, and only the *curve* is contractual (section 5).

The design constraint on the correction kernels is sharp:

Heuristic

A shoulder is a *local* feature: it changes the body of the smile in a bounded region but must not alter the asymptotic wings (those are pinned by liquid risk-reversals and by whatever tail law the base commits to). So the correction kernels must vanish — in value, slope, *and* curvature — in both tails. That "zero-wing" requirement does not *uniquely* select a kernel (a Gaussian and its first two derivatives also vanish in the tails); the centred-second-difference hat of section 3 is the convenient implementation choice: it is built from the *same* log-cosh primitive as the base — so the model and its Jacobian share one set of analytic jets — it is unit-normalized at its centre (α reads directly as the variance displacement there), and its plateau width and shoulder steepness are separate dials (h, κ) where a Gaussian has only one.

2 The sigmoid base

2.1 Coordinates and the log-cosh primitive

Work in the dimensionless log-strike

$$z = \frac{k}{\sigma_{\text{ref}}\sqrt{t}}, \tag{1}$$

where σ_{ref} is the near-the-money quoted volatility, so that z measures moneyness in standard deviations and the model is roughly scale-free across expiries. The convexity primitive is the *log-cosh*

$$\Phi_\kappa(u) = \frac{4}{\kappa^2}\log\cosh\Big(\frac{\kappa u}{2}\Big), \qquad \Phi'_\kappa(u) = \frac{2}{\kappa}\tanh\Big(\frac{\kappa u}{2}\Big), \qquad \Phi''_\kappa(u) = \operatorname{sech}^2\Big(\frac{\kappa u}{2}\Big). \tag{2}$$

Φ_κ is a smooth, convex, even softened absolute value: $\Phi_\kappa(u) \sim \frac{2}{\kappa}|u| - \frac{4}{\kappa^2}\log 2$ for large $|u|$ (a wing of slope $2/\kappa$) and $\Phi_\kappa(u) \approx u^2/2$ near 0 (the rounded vertex), with κ the transition steepness. For numerical safety the log-cosh is replaced by its asymptote $|x| - \log 2$ beyond $|x| = 50$ to avoid cosh overflow.

2.2 The six base parameters

The base slice and its z-derivatives are

$$\begin{aligned} v_{\text{base}}(z) &= V_0 + S_0(z - z_0) + K_0\,\Phi_{\kappa_i}(z - z_0), \\ v'_{\text{base}}(z) &= S_0 + K_0\,\Phi'_{\kappa_i}(z - z_0), \qquad v''_{\text{base}}(z) = K_0\,\Phi''_{\kappa_i}(z - z_0), \end{aligned} \tag{3}$$

with the steepness switching $\kappa_i = \kappa_P$ for $z < z_0$ and $\kappa_i = \kappa_C$ for $z \geq z_0$. The six parameters are the variance level V_0 *at the fitted centre* z_0, the z-slope S_0 there, the convexity amplitude $K_0 \geq 0$, the centre z_0, and the *asymmetric* put/call wing steepnesses κ_P, κ_C. Be precise about what V_0, S_0 are *not*: the centre z_0 is fitted and generally $\neq 0$, so they are not the ATM level and skew, and the hats can move the full model's ATM handles besides — the displayed ATM level/skew/curvature always come from the full curve. Because Φ and Φ' both vanish at the origin, the slice is C^2 across z_0 even though the steepness switches there — so the asymmetry is in the *rate* at which each wing straightens, not a kink. The asymptotic z-space variance wing slopes are

$$v'_{\text{base}}(z) \to S_0 \mp \frac{2K_0}{\kappa_{P,C}} \quad (z \to \mp\infty), \tag{4}$$

i.e. a left slope $S_0 - 2K_0/\kappa_P$ and a right slope $S_0 + 2K_0/\kappa_C$. These are the only handles that set the wings, a fact section 3 exploits.

3 Zero-wing hats

3.1 Definition

The correction kernel is the normalized centred second difference of the log-cosh primitive,

$$B_{c,h,\kappa}(z) = \frac{\Phi_\kappa(u-h) - 2\Phi_\kappa(u) + \Phi_\kappa(u+h)}{2\Phi_\kappa(h)}, \qquad u = z - c, \tag{5}$$

centred at c, of half-width h and steepness κ, normalized so that $B_{c,h,\kappa}(c) = 1$. Its derivatives are the same second difference of Φ' and Φ'' over the same normalizer.

Lemma 1 (Zero wings)**.** $B_{c,h,\kappa}(z) \to 0$, $B'_{c,h,\kappa}(z) \to 0$ *and* $B''_{c,h,\kappa}(z) \to 0$ *as* $z \to \pm\infty$.

Proof. For large $|u|$, $\Phi_\kappa(u) = \frac{2}{\kappa}|u| - \frac{4}{\kappa^2}\log 2 + o(1)$ is asymptotically *affine*, so the centred second difference $\Phi_\kappa(u-h) - 2\Phi_\kappa(u) + \Phi_\kappa(u+h) \to 0$ (a second difference annihilates affine functions). The same holds for Φ' (which tends to the constants $\pm 2/\kappa$) and Φ'' (which decays). Hence B and its first two derivatives vanish in both tails. □

Lemma 1 is the whole point: adding αB to the base changes the body but leaves the wing slopes (4) *exactly* unchanged. The amplitude α is in variance units and, since $B(c) = 1$, is approximately the variance displacement at the centre. A positive α raises a shoulder; a negative α digs a notch. The curvature at the centre is $B''(c) = 2(\operatorname{sech}^2(\kappa h/2) - 1)/(2\Phi_\kappa(h)) < 0$, so a positive hat is locally concave (a hump), as it should be.

The two objects read directly as (2)–(5):

Listing 1: The primitive Φ_κ and the zero-wing hat $B_{c,h,\kappa}$ — the production tail handling verbatim (`models/sigmoid/kernels.py`): past $|x| = 50$ the log-cosh SWITCHES to its exact linear asymptote $|x| - \log 2$; clipping x instead would freeze Φ constant in the far tail and destroy the wing slopes.

```
def safe_logcosh(x):
    # log(cosh(x)) = |x| - log 2 + O(e^{-2|x|})
    x = np.abs(x)
    return np.where(x < 50.0,
                    np.log(np.cosh(np.minimum(x, 50.0))),
```

```
                    x - np.log(2.0))

def phi(u, kappa):          # log-cosh primitive Phi_kappa
    return (4 / kappa**2) * safe_logcosh(kappa * u / 2)

def hat(z, c, h, kappa):    # zero-wing kernel B_{c,h,kappa}
    u = z - c               # centred 2nd difference of Phi:
    raw = phi(u-h, kappa) - 2*phi(u, kappa) + phi(u+h, kappa)
    return raw / (2 * phi(h, kappa))    # so B(c) = 1
```

3.2 The full model

The Multi-Core Sigmoid (MCS) slice is the base plus R signed hats,

$$\boxed{v_R(z) = v_{\text{base}}(z) + \sum_{r=1}^{R} \alpha_r \, B_{c_r,h_r,\kappa_r}(z),} \qquad w(k) = t\, v_R(z), \tag{6}$$

with $6 + 4R$ parameters. By lemma 1 every member of this family has the *same asymptotic wings as its base*, regardless of R — so adding expressiveness in the body never *moves* the tails. Wing-neutral is weaker than Lee-compatible, and the distinction matters: nothing constrains the preserved base slopes themselves. In total-variance k-space the wing slopes are

$$\beta_P = \frac{\sqrt{\tau}}{\sigma_{\text{ref}}}\Big(\frac{2K_0}{\kappa_P} - S_0\Big), \qquad \beta_C = \frac{\sqrt{\tau}}{\sigma_{\text{ref}}}\Big(S_0 + \frac{2K_0}{\kappa_C}\Big), \tag{7}$$

and admissibility requires $0 \le \beta_P, \beta_C \le 2$ (upward wings, under Lee's cap). The calibration bounds of section 5 do *not* enforce this — the hats are asymptotic-wing-neutral, and admissibility of the preserved base wings must be checked separately (it is diagnosed, not guaranteed; the cross-model wing treatment is Note 09). The cores count R is the "cores" slider, the analogue of the LQD Legendre order N and the SVI vs LQD vs LV expressiveness dial.

4 Static no-arbitrage

Butterfly-freedom is the Durrleman condition (8), evaluated by converting the z-space variance and its derivatives to total-variance k-space derivatives via $k = \sigma_{\text{ref}}\sqrt{t}\, z$ and $w = tv$:

$$g(k) = \Big(1 - \frac{k\, w'}{2w}\Big)^2 - \frac{w'^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) + \frac{w''}{2} \;\ge 0. \tag{8}$$

The equivalence "$g \ge 0 \iff$ non-negative density" carries conditions worth stating for this model: it presumes the *raw* variance is positive ($v_R > 0$ — signed hats can drive it negative, and there is no positivity penalty in the fit), the slice is smooth, and the wings behave appropriately; and a finite evaluation grid makes g a *diagnostic*, never a global certificate. Positivity and the floor interact subtly in code: pricing floors the variance at 10^{-8}, and the reported diagnostic is the Durrleman functional of that *priced* (floored) curve — where the floor binds the priced slice is locally constant, its derivatives are zero, and `is_butterfly_free` flags the slice as not-free through its separate raw-positivity check. (An earlier revision mixed the floored value with the raw derivatives

— a functional of no curve at all; fixed and test-locked 2026-07-11. The *penalty*-path g inside the calibrator deliberately keeps the raw derivatives: at collapsed-variance grid points the $1/w$ term then explodes negative, acting as a harsh de-facto barrier against hats that drive variance through zero.) Unlike LQD, MCS is *not* arbitrage-free by construction: a large or narrow hat can drive g negative, so g is reported per fit, the calibration bounds (section 5) keep hats in a benign range, and a default-on *put-wing regularizer* (section 4.1) penalizes residual violations. The wings are asymptotic-wing-neutral by construction (lemma 1) — their Lee-admissibility is a separate check, (7); the cross-model wing treatment is Note 09.

4.1 Where the arbitrage lives, and the put-wing regularizer

When cores over-reach, they break (8) not near the money — where quotes are dense and discipline the curvature — but out in the *sparsely quoted wings*, where each hat's local curvature is unconstrained by data. The offline backtest (project `backtest/`) localizes the violation sharply — with its scope stated: across the *audited SIV-3 spike-regime refits* (the population that motivated the cap; not all event or liquid fits) about 64% of the $g < 0$ points sit in the *put* wing (the steep, variance-loaded side of the equity skew) against only $\sim 4\%$ near the money, with the worst point at a median standardized moneyness $z \approx -3.2$. To discipline that region, the joint refine stage carries an optional soft *put-wing Durrleman penalty*. Its grid $\{z_j\}_{j=1}^{M}$ of $M = 49$ points spans the *entire* interval $[z_{\min} - 2,\ z_{\max} + 2]$ — quoted region included, extended $\Delta z = 2$ into the unquoted tails on both sides — and it appends the residual rows

$$r_j^{\text{wing}} = \sqrt{\lambda_j}\,\max(-g(z_j),\,0), \qquad \lambda_j = \Lambda\,\lambda_0\,(1 + \mathbf{1}\{z_j < 0\}), \tag{9}$$

with g the model's analytic Durrleman functional (8) and $\lambda_0 = 10^3$ the base strength. The surfaced dial is $\Lambda = \texttt{sivWingPenaltyPct}/100$ (default 1; 0 disables), and the indicator *doubles* the weight on the put side to match the measured asymmetry. Because $\max(-g, 0) = 0$ wherever the slice is butterfly-free, (9) is *exactly zero on an admissible fit*: a liquid, arbitrage-free name is byte-identical with or without it. The penalty bites only where a hat would otherwise manufacture a wing violation, pulling the fit back to the admissible boundary there rather than everywhere.

> **Heuristic**
>
> The penalty is a *one-sided* hinge on the same g the model already reports. Although its grid covers the quoted region too, it *bites* where the model extrapolates: where quotes are dense they already discipline the curvature, so $g \geq 0$ there and the hinge rows are zero. It does not *guarantee* $g \geq 0$ (a soft penalty trades the violation against the data fit — the ablation below leaves a residual flagged population even with every defence on), but on genuinely arbitraged illiquid wings it shrinks the worst violation by orders of magnitude — the backtest measured a median wing $\min g$ moving from ≈ -7.9 to ≈ -0.02 — at the cost of fitting *less* to the arbitraged de-Americanized quotes that caused the violation, which is the right trade (Note 05 removes that arbitrage at the source; section 7).

5 Calibration

The fit is a three-stage workflow that respects the model's layered structure.

1. **Base fit ($R = 0$).** Fit the six base parameters to the mid quotes under bound constraints, from a data-driven start that reads the ATM variance, the local skew and the local convexity off three interpolated points. This stage is always mid, so its residual is meaningful for the next stage.
2. **Greedy hat seeding.** Compute the base residual in variance space and place each of the R hats at the largest remaining |residual|, masking a neighbourhood after each placement so two hats cannot seed on the same feature. Each seed starts at half-width $h = 0.40$ and steepness $\kappa = 5.0$, with amplitude initialized *from* the local residual and clipped to $[-1, 1]$ variance units.
3. **Joint bounded refine.** Refine all $6+4R$ parameters together under the requested objective (mid or band) with a mild ridge $\sqrt{\varrho}\,\alpha_r$ on the hat amplitudes, $\varrho = 0.01$. The ridge keeps overlapping cores from exploding against each other without biasing a well-determined amplitude.

All positive parameters (K_0, the steepnesses, the hat half-widths and steepnesses) are bound-constrained directly through scipy's trust-region reflective solver; the hat half-width is held in $[0.15, 1.5]$ and the steepness in $[1.0, 12.0]$. The cores count is capped by the quote count,

$$R \le \Big\lfloor \frac{N_{\text{quotes}} - 6}{4} \Big\rfloor, \tag{10}$$

which guards sparse short-dated chains against fitting spurious narrow kernels. Be precise about what this rule does and does not prevent: it caps only the *hats*, so it stops underidentified hats whenever $N_{\text{quotes}} \ge 6$ — but the application's minimum of *five* included quotes per node means the six-parameter base can still be fitted to five quotes (one parameter over the data; the bounds and the data-driven start keep it tame, but it is not an identifiability guarantee). Independently of this data-driven bound, the surfaced `nCores` is *hard-capped at two*, $R \in \{0, 1, 2\}$: the backtest found a third core buys almost nothing out-of-sample (section 7) while manufacturing the put-wing arbitrage of section 4.1, so any incoming setting above two — persisted *or* freshly submitted, the validator runs on every parse — is silently *clamped* (not rejected). Precision matters here: the clamp lives in the *schema* (`FitSettings.nCores`), not in the calibrator — `calibrate_sigmoid(n_cores=3)` runs happily, and the library's own tests exercise it. The cap is a product decision about what the app will persist and display, enforced at the settings boundary; research code below that boundary deliberately keeps the full family. Relatedly, the hats are *non-unique* by construction (two overlapping hats can trade amplitude against each other): the round-trip test recovers the *curve*, not the parameters, and nothing downstream ever consumes hat parameters directly. The refine stage carries the same optional blocks as the other models — band fit (Note 07), var-swap target (Note 08), the model-agnostic calendar hinge (Note 10), prior persistence (Note 13), the put-wing regularizer (9), and the default-off `extrapEnforce` tapered extrapolated-region fences (Notes 09–10; finite-differenced rows over the analytic core, like SVI) — so the sigmoid overlay is no longer a prior exception.

Performance

The dominant cost of an R-core fit is the $(6 + 4R + 1)$-evaluation finite-difference Jacobian. The closed-form Jacobian of the residual stack (data + ridge + calendar) propagated through the log-cosh primitive replaces it in one pass; a fresh run measured 35.4 ms vs 119.5 ms (3.38×) with the optimum unchanged to $1.5e-15$ in cost. Scope of that number: it is a *final-refine microbenchmark* — the $R = 2$ mid refine alone, clean configuration, on the synthetic WW target — and excludes the base fit, the hat seeding and the product-default wing penalty; it

is not an end-to-end calibration timing. The analytic path is gated to the var-swap/prior-free configuration, as in LQD and SVI. When the put-wing regularizer (9) or the `extrapEnforce` fences are active the Jacobian is *hybrid*: analytic for the data/ridge/calendar rows and finite-difference only for the small penalty blocks, so the penalized fit keeps most of the analytic speed. See appendix B.

6 Worked example: a WW smile

The target in fig. 1 is a synthetic WW built to be *globally* admissible, not merely clean on a window: its *total variance* is a hyperbolic base plus two Gaussian shoulders, $w(k) = a + b\sqrt{k^2+\varsigma^2} + \sum_{\pm} A\, e^{-((k\mp c)/s)^2}$. Gaussians vanish with all their derivatives, so the w-wings are *exactly linear* with slope $\beta = b = 0.055$ — Lee-admissible for every k — and the Durrleman functional tends to the positive limit $(4-\beta^2)/16 = 0.250$ in both tails. The generator asserts $g > 0$ for the target *and* the fitted slice on a dense grid out to $|k| = 12$ before writing anything, which together with the tail limits makes the cleanliness claim global (see the remark closing this section; the shape still has the two interior local maxima proposition 1 forbids a hyperbola). The sigmoid base ($R = 0$) fits it only to 153.05 vol bps, missing both shoulders. Two hats ($R = 2$, $6+4\cdot 2 = 14$ parameters) reduce the maximum error to 2.55 vol bps while preserving the wing slopes $(S_0 - 2K_0/\kappa_P,\ S_0 + 2K_0/\kappa_C) = (-0.017,\ 0.017)$ and keeping the fit butterfly-clean globally (fig. 2(b) plots g on the display window; the wide-grid minimum is 0.28). (How many cores a WW needs depends on the target: this note's WW resolves at $R = 2$, while the library's own, sharper synthetic requires $R = 3$ — which the uncapped library test exercises, below the schema clamp of section 5.) Figure 2 shows the decomposition: the convex base owns the V, and the two signed hats add precisely the two shoulders, each localized and wing-neutral. This regenerated two-core example carries its own regression lock (`test_sigmoid.py::test_note03_ww_two_core_example_regression`).

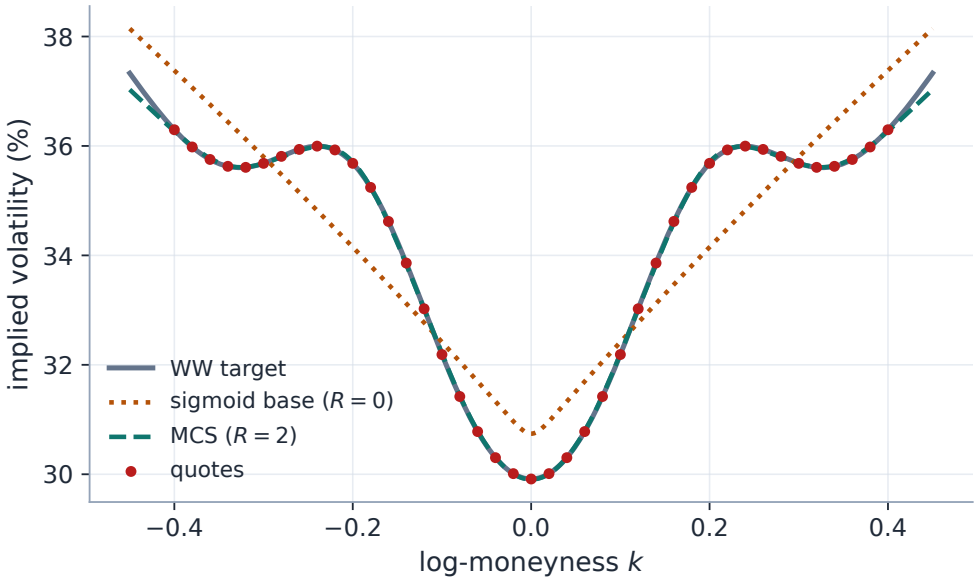
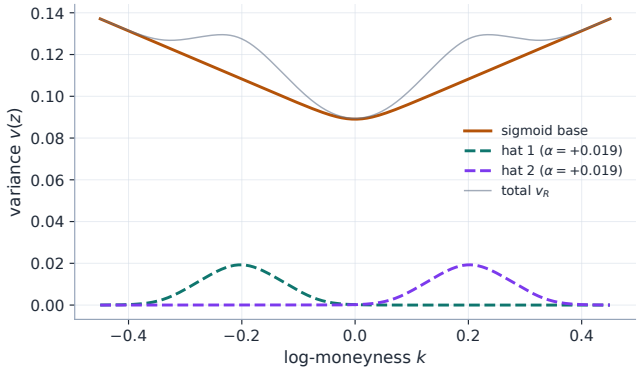
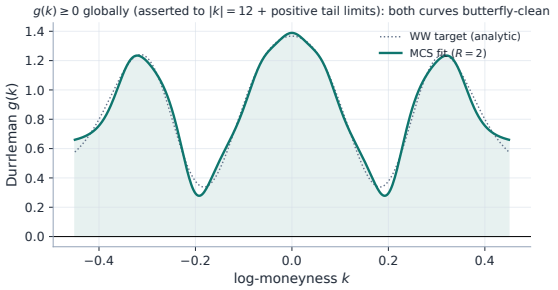


Figure 1: The WW smile: the convex base ($R = 0$, dotted) misses the shoulders by 153.05 vol bps; the two-core MCS fit (dashed) reaches 2.55 vol bps.



(a) base + two signed hats



(b) Durrleman $g(k) \geq 0$, fit and target

Figure 2: Left: the additive decomposition — the convex base owns the wings, the hats own the shoulders. Right: the butterfly diagnostic for both curves — the fitted slice and the synthetic target it chased — non-negative on the display window and, by the wide-grid assertion plus the analytic tail limits, globally.

Remark 1 (The target must be clean too — a bug fixed twice)**.** A diagnostic is only as strong as the exercise it certifies, and this figure's target needed two rounds of honesty. An early revision built a WW target whose *own* Durrleman function dipped to $g \approx -0.38$ near the shoulders: the synthetic "market" itself carried butterfly arbitrage, so no admissible slice could both track it closely and honour $g \geq 0$, and the figure's claim was quietly false. The first fix re-chose the shoulder amplitudes and widths (the shoulder-top curvature scales like $2A/s^2$ and is what drives w'', hence g, negative) — but that target was built in *volatility* space with $\sim 0.20|k|$ tails, so its total variance grew quadratically and still violated Lee far outside the plotted window: the cleanliness was only per-window. The current construction builds the target in *total-variance* space — hyperbolic base + Gaussian shoulders — so the w-wings are exactly linear ($\beta = 0.055 \leq 2$) and the tail g limit $(4 - \beta^2)/16$ is positive analytically; the generator computes g from analytic jets (no finite differences, the measurement lesson of Note 02) and asserts $g > 0$ for target *and* fit on a dense grid to $|k| = 12$ before any figure is written. Current wide-grid values: target $\min g = 0.25$, fitted slice $\min g = 0.28$ (`figures/gen_siv.py`; locked by `test_sigmoid.py::test_note03_ww_target_globally_clean`).

7 Original ideas, and the over-fit caution

The genuinely original elements are the *zero-wing hat* (5)–lemma 1 — a correction basis that is local in the body and provably neutral in the wings, so expressiveness and tail control are decoupled — and the *layered three-stage calibration* that places the hats on residuals rather than optimizing them blind. Together they let one model span from a clean index smile ($R = 0$ — the sigmoid base itself, a different convex family from SVI, not an SVI equivalent) to an event-shaped WW smile ($R = 2$) on a single slider. (A WW *implied-volatility* curve suggests, but does not by itself prove, a bimodal risk-neutral density — the density is the Breeden–Litzenberger second derivative of price, not a re-reading of the vol curve.)

Caution

With great expressiveness comes over-fitting. The offline backtest (project `backtest/`) found that on liquid index chains the Multi-Core hats *over-fit*: cores chase quote noise and manufacture butterfly-arbitrage (the analytic diagnostic flags $g < 0$ on a majority of calibrated nodes at $R \geq 2$ on dense data), while the base $R = 0$ behaves comparably to SVI; the extra cores buy in-sample error and give most of it back out-of-sample. The honest marginal number for the third core, from the spike replay: OOS RMS 13.99 → 13.58 bp — a 0.41 bp improvement — for roughly four times the historical fit cost (514 → 2023 ms in that harness) and worse arbitrage. Not "negative precision", but nowhere near worth it. Two measures followed. The amplitude ridge, the identifiability cap (10), and the g diagnostic mitigate the fit itself; the surfaced slider is *hard-capped at two* (section 5); and the put-wing regularizer (9) of section 4.1 is default-on to catch the residual wing violations that remain within the cap. The operational guidance is unchanged: *use cores only when the smile is genuinely non-convex* (events, bimodal positioning), and prefer LQD or Local-Vol for routine fitting. The cores slider is a scalpel, not a default.

Remark 2 (Two defences, one pathology — and they are complementary)**.** The put-wing violation has *two* sources: the flexible MCS genuinely over-reaching (cured by (9)), and *arbitraged inputs* — the per-strike de-Americanization of Note 05 can hand every model a call curve that is already slightly non-convex in the wings. An ablation isolating the two defences (the de-Am convex repair of Note 05 versus the penalty (9)) on illiquid American ETFs found them *complementary, not redundant*. On the arb-prone population (medians over 38 nodes): with neither, worst-point $g \approx -30$ and in-sample RMS 92 bp; the de-Am repair alone cuts the violation about threefold *and lowers* the RMS to 25 bp (it removes the arbitraged quotes the model was chasing); the penalty alone nearly eliminates the violation (median $\min g \geq -0.02$) but at 749 bp (it must fight those same quotes); both attain the penalty's arbitrage reduction at 225 bp, because the inputs are cleaned before the constraint is imposed. Reduction, not elimination: even with both defences on, 26% of the illiquid population stays flagged — neither defence *guarantees* $g \geq 0$. One real node makes the mechanism vivid (EFA in the Aug-2024 vol spike, 11 days to expiry, 22 quotes): neither defence, 75 bp with worst $g = -116$; repair alone, a tight 18 bp but $g = -12$ remains; penalty alone, arbitrage-free but 726 bp; both, arbitrage-free at 34 bp. The fence is expensive only when it must fight arbitraged inputs — the repair is what makes it affordable. On liquid names the two defences behave differently, and only one is a no-op: the de-Am *repair* never fires there (dense chains have convex de-Am wings — byte-identical, confirmed on real data), but on the *flagged* liquid subset (17 of 30 audited nodes) the *penalty* genuinely trades fit for cleanliness: median in-sample error 10.9 → 36.9 bp with 41% still flagged. That trade is the price of the fence where the violations are shallow ($\min g \approx -0.23$) and the quotes dense. The quantitative case for shipping both on by default stands on the illiquid population; the census that sited the penalty (64% put wing on the audited SIV-3 spike refits, median worst violation at $z \approx -3.2$) and the cross-model framing are Note 09's case file and confinement principle.

8 Traceability

A scoping word first: the "zero-wing / diagnostics" rows below anchor Note 03's *current* construction; the legacy three-core golden tests (`test_note_table1...` etc., from the superseded design note's sharper synthetic) lock the library's uncapped $R = 3$ behaviour and are kept as regression tripwires,

not as evidence for this note's figures — the regenerated two-core example now has its *own* lock. The penalty rows are locked with stated tolerances: penalty-off is byte-identical (`assert_array_equal`); penalty-on over a clean slice is unchanged to 10^{-9} relative; and the repair test accepts a repaired $\min g \geq -0.05$, not exactly zero.

Table 1: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor *Test anchors*
Hat is unit-height, flat at centre, zero-wing in value/slope/curvature	lemma 1	`models/sigmoid/kernels.py` `test_sigmoid.py::test_hat_is_zero_wing_and_unit_height`
Wing slopes preserved under cores; g and $\min v$ diagnostics match note	(4)	`models/sigmoid/sigmoid.py` `test_sigmoid.py::test_note_arbitrage_diagnostics`
Reported g is the priced (floored) curve's functional; degenerate slices flagged not-free	section 4	`models/sigmoid/sigmoid.py::gatheral_g` `test_sigmoid.py::test_floored_diagnostic_is_priced_curve_functional`

This note's two-core WW example (fit tight, base misses shoulders; target and fit globally butterfly-clean)	section 6	`models/sigmoid/calibrate.py`
		`test_sigmoid.py::test_note03_ww_two_core_example_regression`
		`::test_note03_ww_target_globally_clean`

Cores improve the WW fit monotonically (legacy synthetic)	section 6	`models/sigmoid/calibrate.py` `test_sigmoid.py::test_more_cores_fit_better_on_ww_smile`
Hat-count cap $R \leq \lfloor (N-6)/4 \rfloor$ (hats only; the base still fits at $N = 5$)	(10)	`models/sigmoid/calibrate.py` `test_sigmoid.py::test_cores_are_capped_to_quote_count`
Surfaced slider clamped to two (schema boundary, every parse)	section 5	`api/schemas.py` `test_siv_wing_penalty.py::test_cores_clamped_to_two`

Put-wing hinge repairs wing arb (to $\min g \geq -0.05$); clean slices unchanged (10^{-9} rel); off by library default (byte-identical)	(9)	`models/sigmoid/calibrate.py`
		`test_siv_wing_penalty.py::test_penalty_removes_wing_arb`
		`::test_penalty_byte_identical_on_arb_free_slice`
		`::test_penalty_off_default`

Analytic Jacobian matches FD (base, cores, band, calendar)	section 5	`models/sigmoid/jacobian.py`
		`test_sigmoid_jacobian.py::test_two_cores_mid`
		`::test_two_cores_band`
		`::test_calendar_active`

Curve identified, parameters deliberately not	section 5	`models/sigmoid/calibrate.py` `test_sigmoid.py::test_round_trip_curve_recovery`

Claim	Object	Code anchor *Test anchors*
Prior blocks reach the MCS overlay	section 5	`models/sigmoid/calibrate.py` `test_prior_parametric.py::test_operator_prior_pulls_all_models_toward_prior_skew`
R3×R6 ablation numbers	section 7	`backend/backtest/ablation_arb.py` `test_ablation_arb.py`
WW figure target and fit butterfly-clean globally (dense grid to $\|k\| = 12$ + analytic tail limits)	remark 1	`figures/gen_siv.py` runtime assertions in the generator + the two example tests above

A Hyperparameter atlas

Table 2: Multi-Core Sigmoid (MCS) hyperparameters.

Knob		Default	Role
Surfaced. Product defaults shown (the product's default model is LQD; these apply when MCS is selected). Library defaults differ: $R = 0$, penalty off.			
`nCores` (`FitSettings`)	R	2	Number of zero-wing hats, $R \in \{0, 1, 2\}$ (schema-clamped at two on *every* parse — persisted or fresh; the library itself is uncapped, and the identifiability bound (10) may reduce R further). 0 = pure sigmoid base.
`sivWingPenaltyPct` (`OptionsSettings`)	Λ	100	Put-wing Durrleman penalty (9) strength; 100 = base $\lambda_0 = 10^3$, 0 = off.
`sigmoidRidge` (`FitSettings`)	ϱ	0.01	ℓ^2 penalty on hat amplitudes in the refine stage.
`midAnchorWeight`		0.05	Band-fit mid anchor (Note 07).
`weightScheme`		`equal`	Per-quote weights (Note 07).
`calendarWeight`		10^6	Calendar hinge penalty (Note 10; `OptionsSettings`).
`extrapEnforce`		off	Tapered extrapolated-region fences (Notes 09–10; `OptionsSettings`); hybrid Jacobian when on.
Internal (seeding / bounds)			
`_H_INIT`		0.40	Hat half-width seed.
`_KAPPA_INIT`		5.0	Hat steepness seed.
`_H_BOUNDS`		$[0.15, 1.5]$	Hat half-width bounds.
`_KAPPA_BOUNDS`		$[1.0, 12.0]$	Hat steepness bounds.
`_C_PAD`		0.5	Centre padding beyond the quoted z-range.
`_V_FLOOR`		10^{-8}	Variance floor so $\sqrt{v}$ stays real.
`_LOGCOSH_CLIP`		50	$\|x\|$ beyond which $\log\cosh$ uses its asymptote.
`xtol/ftol`		10^{-12}	trf tolerances.

B Performance notes

1. **Analytic Jacobian** through the log-cosh primitive replaces the $(6 + 4R + 1)$-evaluation finite difference; measured 3.38× on the *final-refine microbenchmark* (the $R = 2$ mid refine alone — base fit, seeding and the product-default wing penalty excluded; single-threaded development machine, fastest of three), optimum unchanged to $1.5e - 15$. Gated to the var-swap/prior-free path. When the put-wing penalty (9) or `extrapEnforce` is active, only the small penalty grids are finite-differenced (a *hybrid* Jacobian), so the penalized fit retains most of this speed.

2. **Layered warm start.** Fitting the base first and seeding hats on its residual gives the joint refine an excellent start, so the bounded trf converges quickly even with several cores.

3. **Vectorized kernels.** Φ, Φ', Φ'' and the hat second differences are NumPy-vectorized and side-effect free, shared between the model and its Jacobian.

4. **Identifiability cap.** Limiting R by the quote count avoids wasting iterations on under-determined narrow kernels and is the first line of defence against the over-fit of section 7.

C Reference implementation

Building on listing 1, the full slice $v_R(z)$ (6) and the butterfly diagnostic $g(k)$ (8) are a few more lines. The z-jets (value, slope, curvature) are carried together because g needs all three; `_dhat` forms a hat or either of its derivatives from the matching Φ-jet, so the zero-wing second difference is written once. Verified against `MultiCoreSiv` on a two-core benchmark to 10^{-13}.

Listing 2: The full model $v_R(z)$ and Durrleman $g(k)$ (distilled from `models/sigmoid/{kernels,sigmoid}.py`).

```
def phi_p(u, k):  return (2/k) * np.tanh(k*u/2)          # Phi'
def phi_pp(u, k): return 1 / np.cosh(k*u/2)**2           # Phi''
def _dhat(f, z, c, h, kappa):                            # B or a B-
    derivative
    u = z - c                                            #   from the
        matching Phi-jet
    return (f(u-h,kappa) - 2*f(u,kappa) + f(u+h,kappa)) / (2*phi(h,
        kappa))

def v_model(z, base, cores):
    """MCS variance v_R(z) = v_base(z) + sum_r alpha_r B_r(z), with z-
        jets."""
    v0, s0, k0, z0, kp, kc = base
    u = z - z0
    kappa = np.where(u < 0, kp, kc)                      # asymmetric
        wings, C^2 at z0
    v, vz, vzz = v0 + s0*u + k0*phi(u,kappa), s0 + k0*phi_p(u,kappa),
        k0*phi_pp(u,kappa)
    for alpha, c, h, kap in cores:                       # add the
        signed zero-wing hats
        v   += alpha * _dhat(phi,    z, c, h, kap)
        vz  += alpha * _dhat(phi_p,  z, c, h, kap)
        vzz += alpha * _dhat(phi_pp, z, c, h, kap)
```

```
    return v, vz, vzz

def durrleman_g(z, base, cores, t, sigma_ref):
    """Butterfly diagnostic g(k) >= 0; converts z-space jets to k-space."""
    v, vz, vzz = v_model(z, base, cores)
    k, w = sigma_ref*np.sqrt(t)*z, t*v
    wk, wkk = np.sqrt(t)/sigma_ref*vz, vzz/sigma_ref**2
    return (1 - k*wk/(2*w))**2 - 0.25*wk**2*(1/w + 0.25) + 0.5*wkk
```

References

[1] J. Gatheral. *The Volatility Surface: A Practitioner's Guide*. Wiley, 2006.

[2] V. Durrleman. From implied to spot volatilities. *Finance and Stochastics*, 14(2):157–177, 2010.

[3] R. W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004.